Current and Pending (Other) Support

Edward A. Varvak

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Proposals and active projects

1. Postdoctoral Fellowship: MSPRF: The Hodge theory of Lagrangian fibrations, with applications to hyper-Kähler geometry

Status of support:	Pending
Proposal/award number:	Leave blank (assigned at submission)
Source of support:	National Science Foundation (Mathematical Sciences Postdoctoral Research Fellowships, NSF 23-603)
Primary place of performance:	Massachusetts Institute of Technology, Cambridge, MA, USA
Project/proposal start date:	09/2027
Project/proposal end date:	08/2031
Total anticipated amount:	$190,000
Person-months per year:	2027: 8.00; 2028: 8.00; 2029: 8.00; 2030: 0.00

Overall objectives. The project studies the Hodge theory of Lagrangian fibrations of hyper-Kähler manifolds and related holomorphic symplectic varieties. Its objectives are (1) to determine which decompositions of the Mumford–Tate group occur for fibers of Lagrangian fibrations, and whether a decomposable fiber forces a hyper-Kähler manifold to be deformation equivalent to a Hilbert scheme of points on a K3 surface; (2) to extend this Hodge theory to non-compact Lagrangian fibrations, chiefly the Hitchin fibration for SL_r-Higgs bundles, by determining the Mumford–Tate group of its very general fiber; and (3) to work toward the hyper-Kähler Bogomolov–Mumford conjecture by producing rational curves from degenerations of fibers with an elliptic factor.

Statement of potential overlap. None. This is the proposal under consideration, and the PI has no other current or pending support.

In-kind contributions

None.

Certification

Completed in SciENcv when the document is generated: the PI certifies that the information on current, pending, and other support (both foreign and domestic) is current, accurate, and complete.